

Ponds of Mercy - Can urban wetlands survive and flourish in Sri Lanka?

A preliminary discussion about the conservation and development of the Bellanwilla and Bandaragama Urban Wetlands



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Summary

- In the 13th Century the Chinese began to refer to wetlands and small lakes as “Ponds of Mercy”
- Wetlands are almost always targeted for urban expansion since they are considered wastelands.
- Global research and observations show the numerous impacts of urbanization on the hydrology, geomorphology, ecology, biodiversity, water quality, water quantity regulation and socio-economics of wetlands
- The Western Province of Sri Lanka has seen the most development since independence which has drastically affected the vast wetlands in the area reducing them to scattered remnant patches

Study Sites

- Bellanwillia-Attidiya (BWA) Wetland in the Dehiwala/Ratmalana Municipality.
- Bandaragama Wetlands (BNG) in the Bandaragama and Kalutara Divisional Secretary Divisions



Study Sites

Bellanwilla-At
tidiya (BWA)
Wetland in the
Dehiwala /
Ratmalana
Municipality.

Study Sites

- Bandaragama
Wetlands (BNG) in
the Bandaragama
and Kalutara
Divisional Secretary
Divisions

Methodology and Materials Used

Ecology

- This study was conducted between August 2005 and December 2005 to ensure that the sample will capture both wet and dry seasons as well the migrant and non-migrant season birds.
- Initially a reconnaissance survey was carried out within the declared boundary of the sanctuary to identify the type of habitats present.
- Five main habitat types were identified based on the reconnaissance survey.
- The general distribution of the five habitats was marked on a 1:50000 map using a hand held GPS.
- Within each habitat random line transect surveys were conducted to identify the type of fauna and flora present.

Ecological Findings @ BWA

-Floral-

- Important Records

- This ecosystem is going through a natural succession and most of the existing habitats are in a state of flux
- Of the five habitats the main plant type recorded in the home gardens were trees most of which were introduced species with an economic value such as timber or fruit trees
- The species recorded did not include any endemic or threatened species

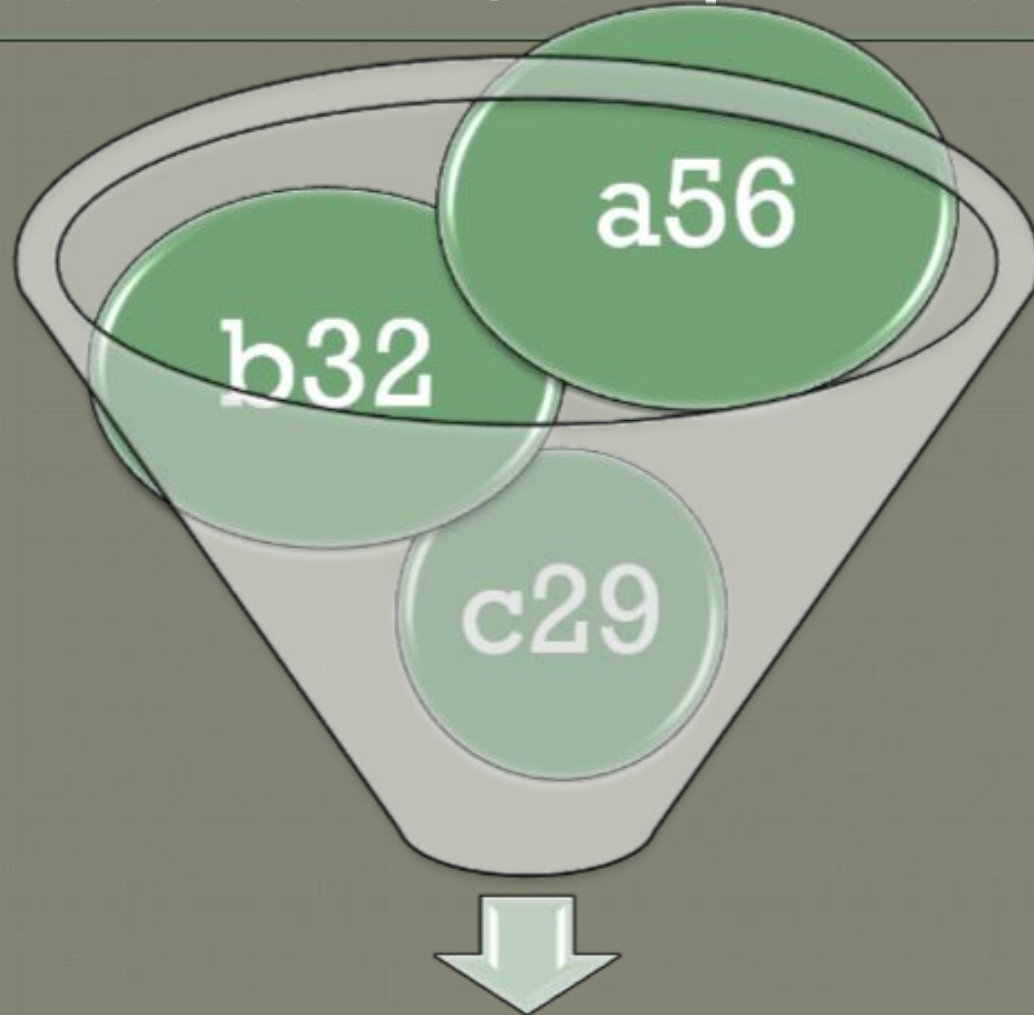
Type of vegetation	Native	Introduced
• Trees	• 14	• 15
• Shrubs	• 17	• 4
• Herbs	• 48	• 19
• Climbers	• 13	• 7
• Epiphytes	• 1	• -

5 Main Vegetation Types

- Marshland vegetation
- Annona dominated woodland vegetation
- Open water vegetation
- Canal bank vegetation
- Home garden vegetation

Ecological Findings @ BWA

-Faunal Historical Comparison-



a= Gunatilleke (1992)

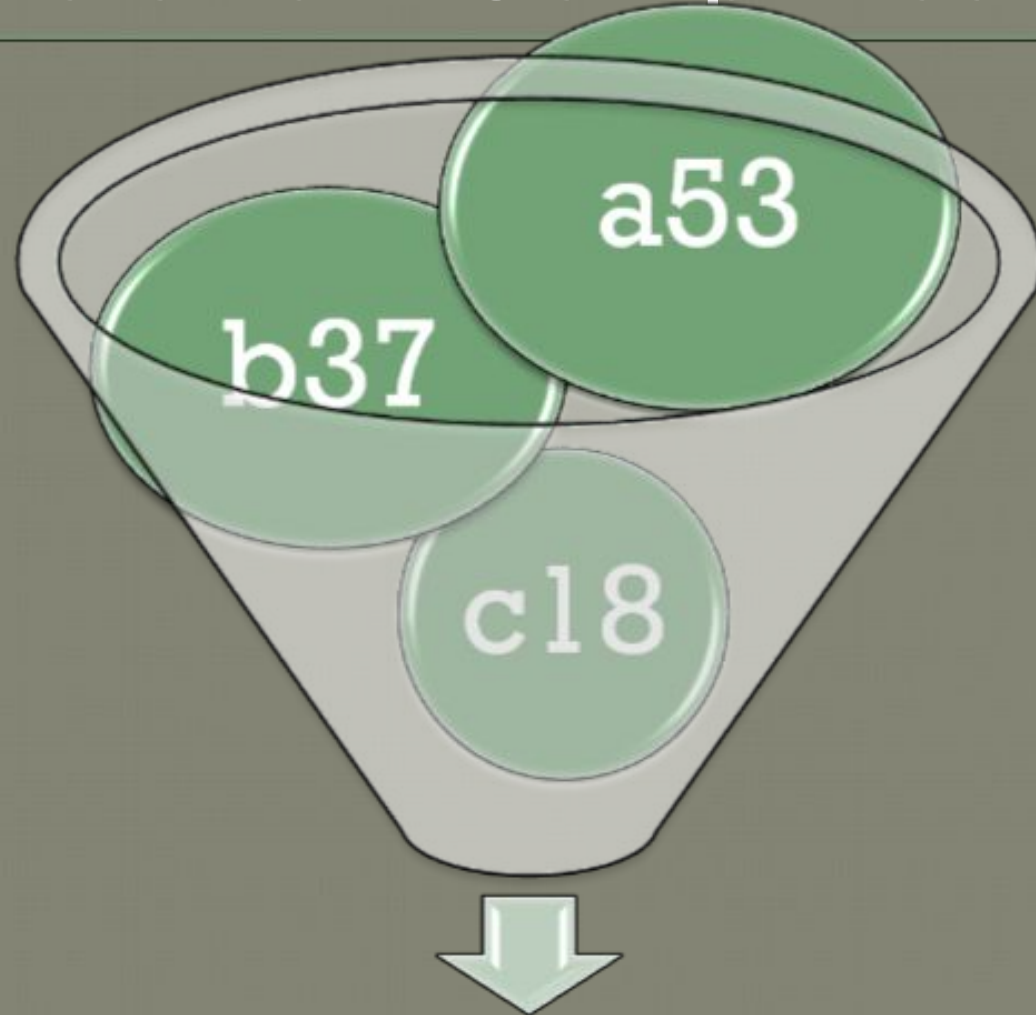
b= Goonatilake (1994)

c= SLWCS Study (2005-2006)

Butterflies

Ecological Findings @ BWA

-Faunal Historical Comparison-



a= Goonatilake (1994);

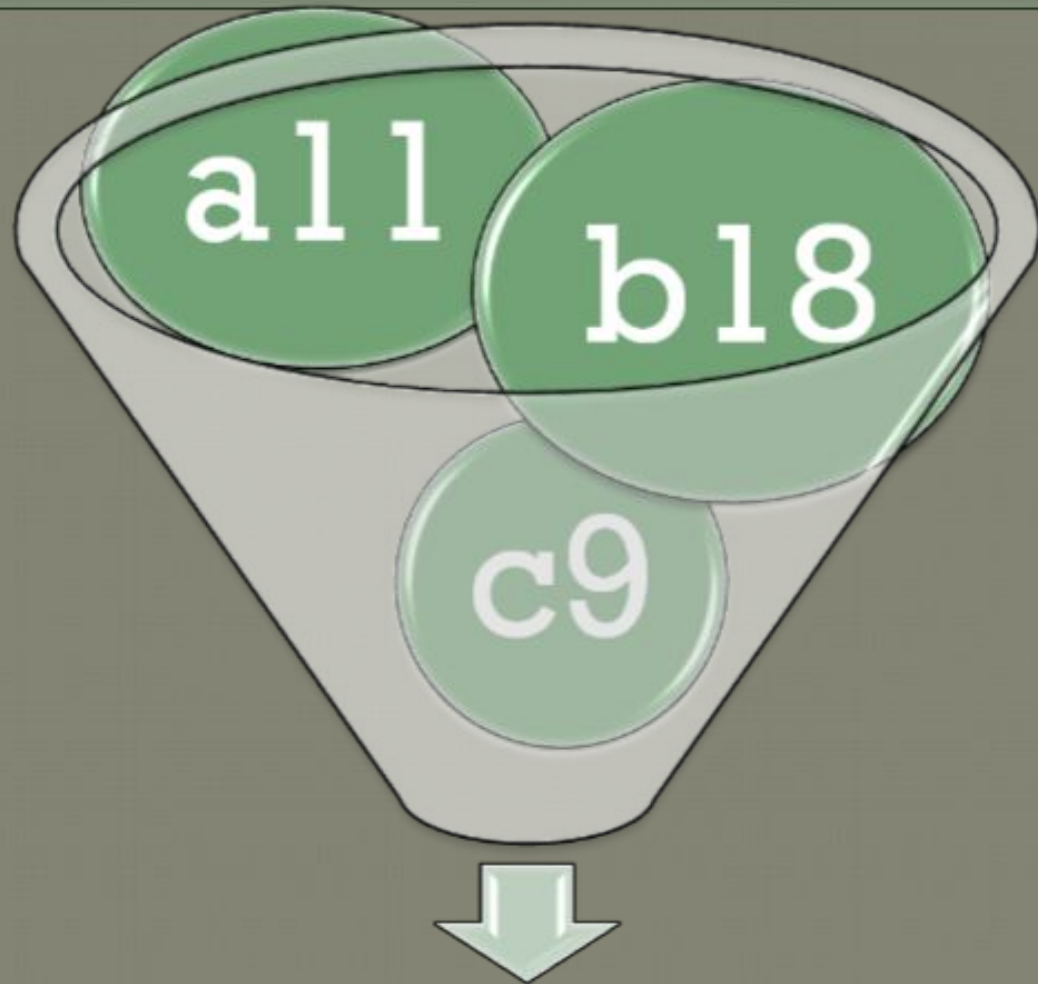
b= Maduranga (2000);

c= SLWCS Study (2005-2006)

Fresh water fish

Ecological Findings @ BWA

-Faunal Historical Comparison-



a= Goonatilake (1994);

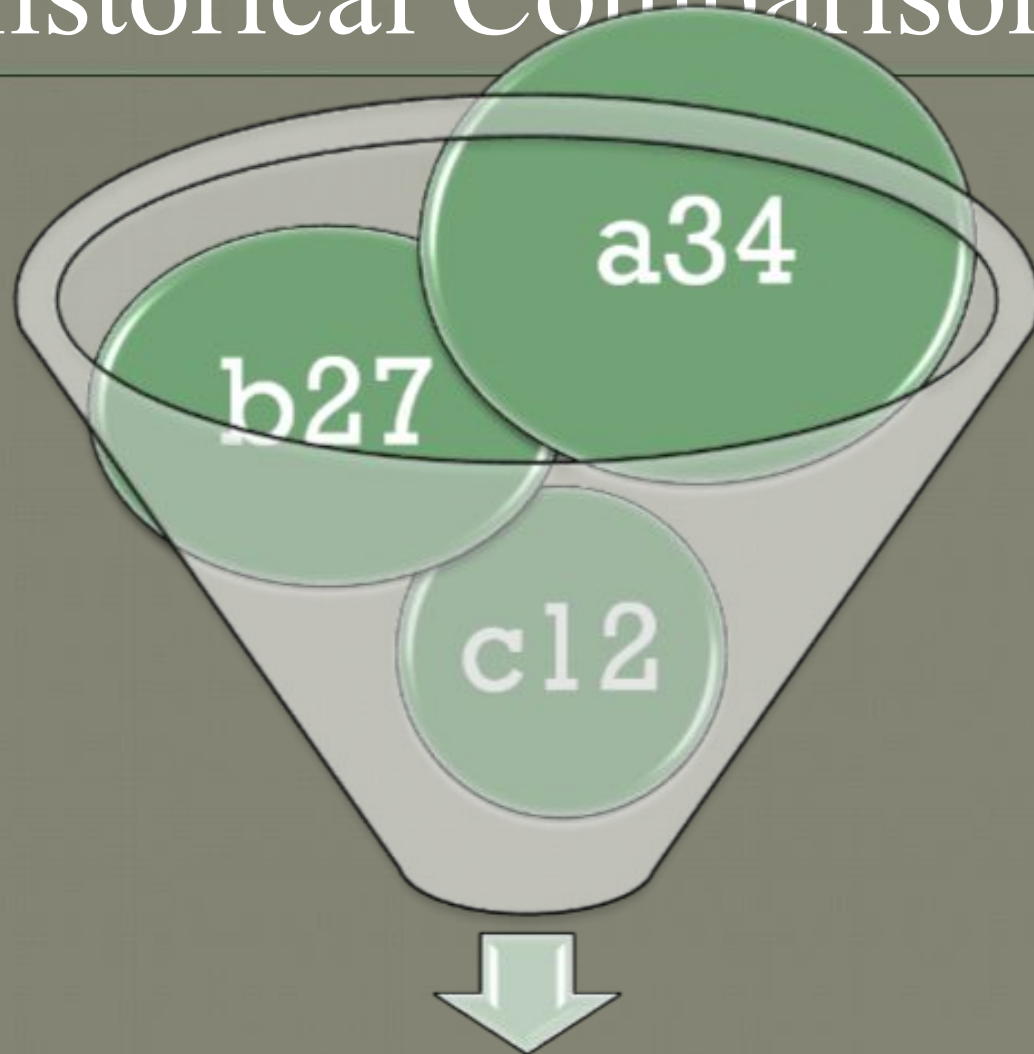
b= Goonatilake (1998)

c= SLWCS Study (2005-2006)

Amphibians

Ecological Findings @ BWA

-Faunal Historical Comparison-



a= Nanayakkara (1988);

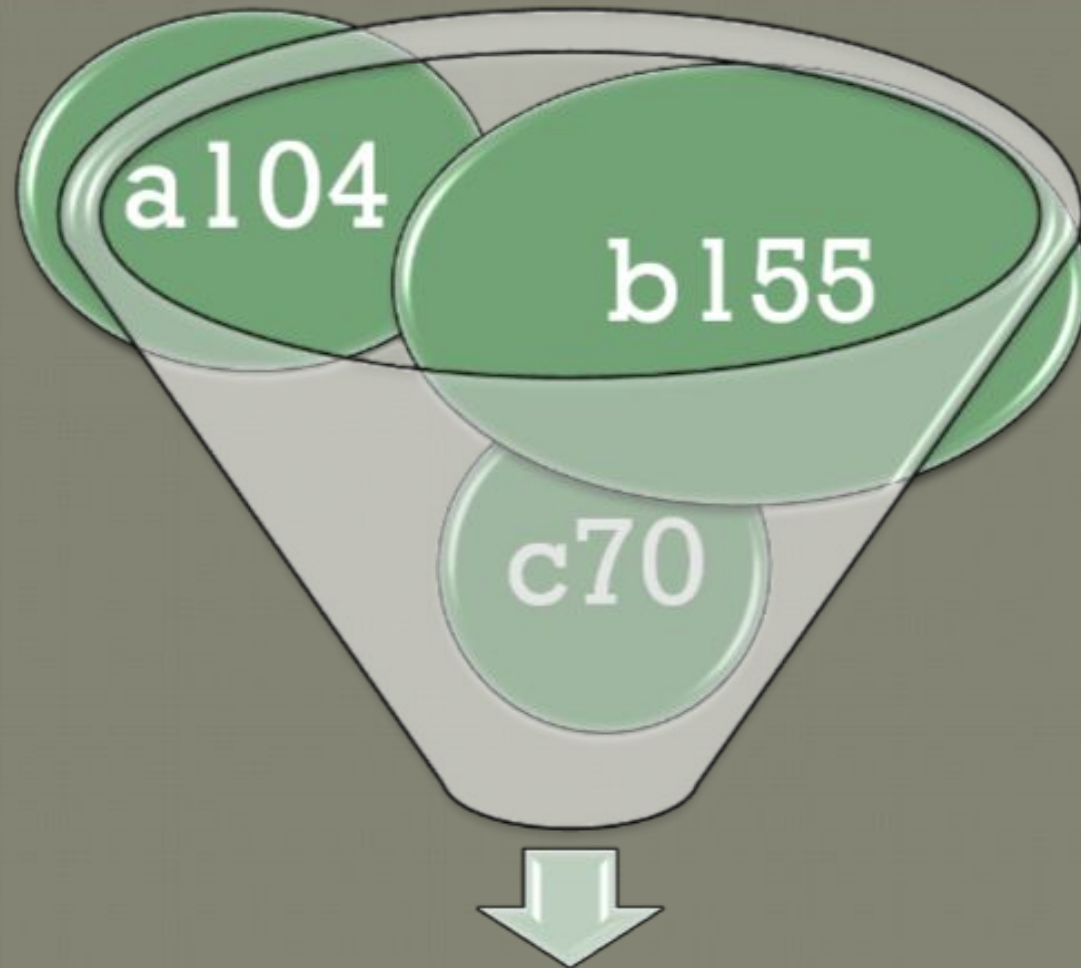
b= Goonatilake (1994)

c= SLWCS Study (2005-2006)

Reptiles

Ecological Findings @ BWA

-Faunal Historical Comparison-



a= Nanayakkara (1988);

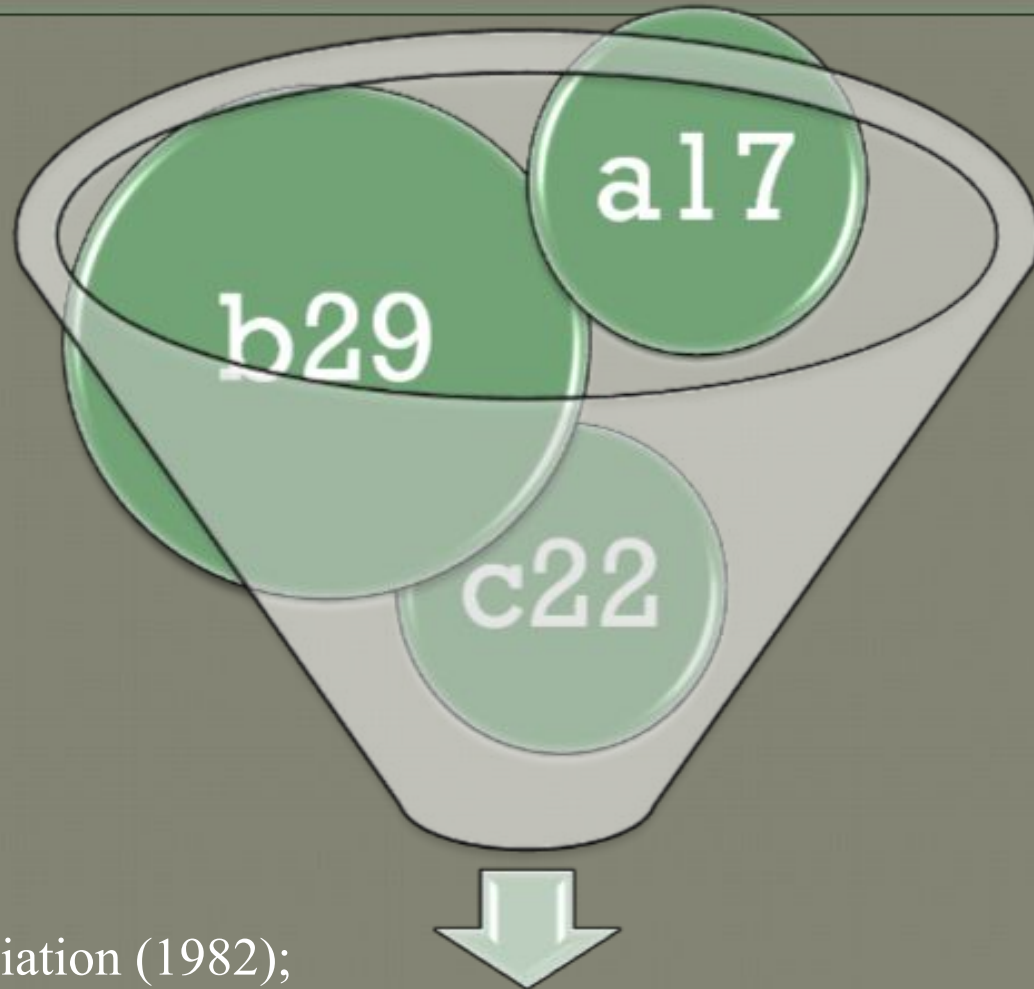
b= Goonatilake (1994)

c= SLWCS Study (2005-2006)

Avifauana

Ecological Findings @ BWA

-Faunal Historical Comparison-



a= Young Zoologist Association (1982);

b= Goonatilake (1994);

c= SLWCS Study (2005-2006)

Mammals

	BWA	BNG
Survey Period	Aug 2005 - Dec 2005	Dec 2005

Number of species observed

	BWA	BNG
Butterfly	29	30
Amphibian	9	7
Reptile	11	13
Mammal	20	10
Bird	62	74
Dragonfly	3	11
Fish	17	12
Plant	138	Not surveyed

Methodology and Materials Used for Looking at Human Impact

Combinations of Rapid Rural Assessment (RRA) and Participatory Rural Assessment (PRA) tools were used.

- Various stakeholders were interviewed, varying from local communities to government representatives and NGO's involved in the area.
- Discussions/meetings with stakeholders (Government and Private)
PRA/RRA tools with communities residing in the area (Venn Diagram, Focus Group Discussions and Problem and Solution Trees)
- Semi-structured Interviews
- Transect Maps of the area

Bellanwillia-Attidiya (BWA)

Data collection: 5 month field survey (Aug to Dec 05) - 10 to 15 days each month were spent in the field.

Number of Interviews (RRA Survey): 79 (40%)

GND's covered: 3

DSD's covered: **1**

Stakeholders met: 13

(Govt. Departments and NGO's)

Community maps and Ven Diagrams were also created.

Bandaragama (BNG)

Data collection: 2 week field survey
(6 to 24 Dec 05)

Number of Interviews (RRA Survey): **n/a**

GND's covered: 5

DSD's covered: 2

Stakeholders met:

(Govt. Departments and NGO's) Agriculture officers in Bandaragama.

Human Impact @ BWA

- Diminishing ecological biodiversity due to:
 - Human use of plants, fishing, hunting, wood in area
 - Waste disposal (industrial effluent, sewage, inorganic refuse) attracting rats, crows, cattle
 - Illegal activities (drugs & manufacture of liquor)
 - Poor canal maintenance
 - Unpleasant odour of area
 - Stagnant and eutrophic water (from waste dumped and poor drainage which causes illness)
 - Uncontrolled settlements and landfilling
 - Poor sanitation (e.g. 2 toilets for 50 families)



Human Impact @ BNG

- Diminishing ecological biodiversity due to:
 - Discharge of human waste into streams (including from slaughter houses)
 - Illegal activities (manufacture of liquor along waterways)
 - Poaching of aquatic birds
 - Pressure of land reclamation and land filling for construction and development
 - Presence of Knife Fish reduce native fish population
 - Poor canal maintenance causing flooding



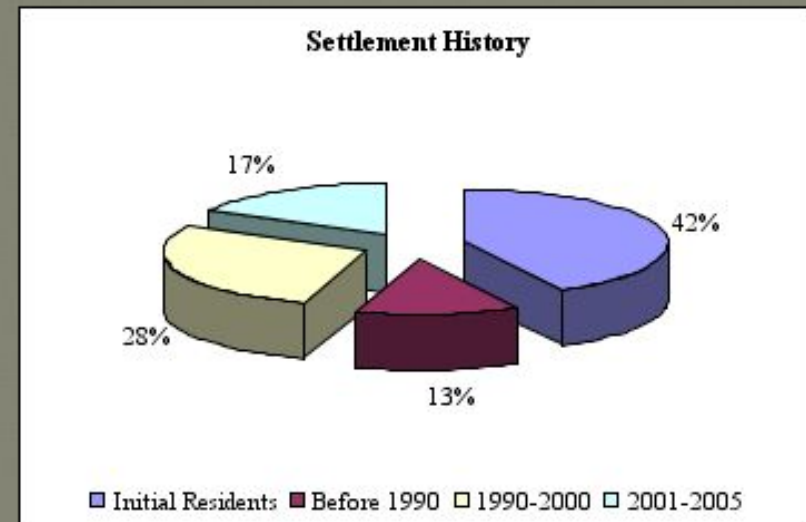
Why does this happen?

- Poor facilities (esp sanitation) for residents
- No government subsidies available
- No settlement control
- Lack of education on conservation and affect of pollution to the environment
- Low income families
- No facilities for solid waste management
- Poor community participation and cooperation

Human Impact @ BWA and BNG

- The project collected primary data on

- Religion
- Ethnicity
- Education
- Access to services (health, electricity, water, sanitation and housing)
- Livelihood systems
- Income fishing practices
- awareness of environmental legislation and fines
- Land tenure
- Gender issues and roles
- Tourism
- The work of Government agencies, CBOs, NGOs and INGOs in the area



DSD	Upland (Rs./Perch)	Lowland (Rs./Perch)
Millaniya	16500	3000
Bandaraga	40900	10800
ma		
Average	28700	6900



Human Impact @ BWA

- Highlight the ecological, biological and socio-economical importance of KKP to local, national, regional and international conservation efforts -

1. COMMON PROBLEMS

- a. Poor facilities for residents in the area
- b. No government subsidies due to settlements living in a sanctuary area
- c. Settlements increasing and no action being taken to control the matter
- d. Today neither well nor marsh water can be used instead piped water has to be purchased to drink

2. HEALTH

- a. Children of families settled along the canal banks have respiratory problems
- b. Mosquito breeding (Malaria/ Dengue)
- c. Poor sanitation problems (e.g. 2 toilets for 50 families)



3. EDUCATION

- a. Lack of awareness about the marshes (e.g. importance, conservation and education on flora and fauna)
- b. No clear demarcation of the sanctuary area

4. INDUSTRIES

- a. Harmful industrial effluent discharged in the canals
- b. Industrial owners have strong political links
- c. Poor rules and regulations to control this problem
- d. Fish are affected by chemicals and cannot be eaten

6. SETTLEMENTS

- a. They feel there is unplanned development taking place (since some of these settlements were moved here under the coastal conservation project i.e. Dehuwala)
- b. No community involvement in formulation of management plans
- c. Political Obstruction
- d. Problems of high human density and illegal settlements
- e. Lack of basic facilities (especially sanitation)
- f. Drugs, alcohol and stealing is more common now
- g. Households get flooded in the rainy season

5. SOLID WASTE MANAGEMENT

- a. Discharge of sewage and other waste into the canals
- b. Lack of finances and proper place for solid waste management
- c. Poor community participation and cooperation
- d. Bad odor from the canals
- e. Increase in crows, flies and mosquitoes creating problems

Methodology and Materials Used

SLWCS Comprehensive Rapid Evaluation System (CRES) is an objective research based priority ranking system that utilizes multiple techniques such as

- DELPHI
- Participatory Rural Appraisals and Rapid Rural Appraisals
- Central Environmental Authority (SL-CEA) Wetland System
- North American Wetlands Conservation Councils –Canada method
- RECCE Method for New Zealand Surveys
- Broad Reconnaissance Survey
- Point Count methods

It takes data at national and provincial levels taking the village (GN level) and community members as the smallest unit of information.

CRES ensures that a site is evaluated based on its natural, socio-economic and cultural values in a way that sites and sub-sites can be compared and actions prioritized.

Discussion

Attributes

BWA

BNG

Biological Diversity

- 168 Species of bird recorded (5 globally threatened)
- 138 Plant Species recorded (herbs & shrubs)
- Marsh area

- Low faunal diversity
- 74 Species of bird recorded

Scenic & Landscape

- Open space for recreation
- Polluted, thus no landscape beauty at present

- No particular attraction at present

Historical & Cultural

- No historical significance known

- No historical significance known

Uniqueness

- Unique recording of globally threatened Spot-Billed Pelican
- No longer a unique ecosystem (due to human impact)

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Urgent Actions

Initiate environmental education programs

- such as the Outdoor Environmental Education Program and the Mobile Environmental Education Workshop to educate on sustainably managing the natural environment

Disseminate of the report and organize forums exhibitions,

Organize workshops and presentations to educate the general public, bureaucrats, and policy makers.

Establish a historic and current land-use change analysis program using GIS and satellite imagery

Conduct meetings with stakeholders

- the regional and national level
- the Delphi technique will be utilized to establish relative importance factors

Strengthen and support the DWC, FD and CEA to actively protect the declared sites while allowing sustainable resource utilization.

Think outside the box

- We need to consider solutions such as
 - Public/Private cooperatives
 - Multi-use conservation/conservation easement
- We need to educate/partner with universities, local and national authorities
- We need to work together -more than just laws and regulations.

Develop management strategies to promote sustainable use of resources that will ensure livelihood security as well as financial stability. These efforts must provide incentives as a way of encouraging communities to give up harmful and

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The full and comprehensive report (along with similar reports for Seruvila-Alai-Ulakalie, Kalpitiya-Karaitive-Puttalam and Flood Plains Wetlands) are published and available at www.slwcs.org/research as part of our “Wetland Conservation Project (WCP)”

Hard copies of SLWCS reports, articles and scientific/nature publications can be referred to and obtained from the library at the Society's head office

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